

IOT-ENABLED RFID FRAMEWORK FOR AUTOMATED HOSPITAL MANAGEMENT.

SDG Goals: SDG 3 → Good Health and Well-being (Main Goal)
 SDG 9 → Industry, Innovation and Infrastructure
 SDG 16 → Peace, Justice and Strong Institutions

Cohort Number: 07

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ABSTRACT

Hospital Management using Radio Frequency Identification (RFID) technology provides a smart and efficient solution to improve identification, access control, and data management in healthcare systems. In this project, RFID cards are assigned to different users such as patients, doctors, staff, and administrators, where each card contains a unique identification number. This unique ID helps in accurately identifying individuals and managing their roles within the hospital environment.

When an RFID card is scanned, the system reads the unique ID and verifies it with the central database. Based on the user type, the system retrieves and displays relevant information such as patient details, medical history, appointment schedules, and doctor information. This process reduces dependency on manual record-keeping, minimizes human errors, and ensures faster access to important data, especially during emergencies.

The system also provides an effective access control mechanism, where only authorized users can access specific data or areas, thereby improving security and privacy within the hospital. Additionally, all activities are recorded and monitored through a dashboard, enabling real-time tracking of user interactions and maintaining proper logs for administrative purposes. This helps hospital management in analyzing data and improving workflow efficiency.

The implementation of this system enhances overall hospital operations by reducing delays, improving accuracy, and ensuring better coordination among patients, doctors, and staff. Although the initial setup may require investment, the long-term advantages such as improved efficiency, better data management, and enhanced patient satisfaction make RFID technology a valuable solution in modern healthcare systems.

Overall, this project demonstrates how RFID-based hospital management systems can create a more secure, reliable, and user-friendly environment, contributing to improved healthcare services and operational effectiveness.

INTRODUCTION

In today's fast-paced healthcare environment, hospitals are expected to provide services that are not only accurate but also quick and efficient. However, many hospitals still rely on traditional methods for managing patients, staff, and records. These manual systems often lead to delays, human errors, and difficulty in accessing important information, which can affect the quality of patient care. As a result, there is a growing need for smart and automated solutions to improve hospital operations.

Radio Frequency Identification (RFID) is a modern technology that allows wireless identification of individuals using unique tags. It has already been successfully used in various fields such as transportation, retail, and logistics. In healthcare, RFID has the potential to simplify processes like patient identification, staff management, and data handling.

This project focuses on developing an RFID-based Hospital Management System that assigns unique RFID cards to patients, doctors, staff, and administrators. When these cards are scanned, the system quickly identifies the user and retrieves relevant information from a central database. This helps in reducing manual work, minimizing errors, and improving the speed of accessing important data.

The system also provides controlled access, ensuring that only authorized users can view or manage specific information. Additionally, a user-friendly dashboard is used to monitor activities and maintain records in real time.

Overall, this project aims to create a more efficient, secure, and reliable hospital management system using RFID technology, ultimately improving hospital workflow and enhancing patient care.

PROBLEM STATEMENT

In many hospitals, the management of patients, staff, and medical records is still carried out using traditional and manual methods. These systems often result in patient misidentification, delays in accessing information, and errors in data handling, which can affect the quality of healthcare services. Additionally, there is a lack of efficient access control, making it difficult to ensure that only authorized personnel can view or modify sensitive information. The absence of a centralized and automated system also leads to inefficient workflow, increased administrative workload, and poor coordination between different departments. In critical situations, the inability to quickly retrieve accurate patient information can lead to delays in treatment and decision-making.

Therefore, there is a need for a smart and reliable system that can automatically identify users, manage data efficiently, and provide secure access control. This project aims to address these challenges by developing an RFID-based Hospital Management System that improves accuracy, reduces manual effort, and enhances overall hospital efficiency.

OBJECTIVES

- Develop an RFID-based identification system for patients, doctors, staff, and administrators to ensure accurate and fast identification.
- Automate hospital management processes by reducing manual work and minimizing human errors.
- Design a centralized database system for storing and retrieving patient and staff information efficiently.
- Implement secure access control, allowing only authorized users to access specific data or system features.
- Provide a user-friendly dashboard for monitoring, managing, and displaying real-time information.
- Improve overall hospital efficiency by enabling quick access to data and better coordination between departments.
- Enhance patient care and service quality through faster response time and accurate information handling.

SCOPE OF THE PROJECT

- Develop an **RFID-based system** for identification of patients, doctors, staff, and administrators
- Enable **quick and accurate retrieval of user information** using RFID scanning
- Provide a **centralized database** for storing and managing hospital data
- Include features like:
 - Patient details
 - Appointment management
 - Doctor information
 - User activity monitoring
- Implement basic access control to allow only authorized users to access data
- Design a user-friendly dashboard for real-time monitoring and data display
- Improve hospital workflow efficiency by reducing manual work and errors
- Suitable for small to medium-scale implementation or prototype model
- Limited to basic functionality and does not include:
 - Full hospital system integration
 - Advanced equipment tracking
 - Large-scale deployment
- Can be **extended in future** with more advanced features and technologies

METHODOLOGY

The development of the RFID-based Hospital Management System was carried out using a structured and modular methodology, ensuring each subsystem functioned independently before being integrated. This method facilitated easier debugging, testing, and refinement throughout the project.

1. System Design Planning

The first step involved identifying the key functions required in the hospital environment such as patient identification, access control for medical staff, equipment access tracking, and prescription data retrieval.

Based on these, appropriate components were selected:

- ESP32 was chosen for its dual functionality of microcontroller and Wi-Fi connectivity.
- RC522 RFID Reader and RFID passive tags (cards/key fobs) were selected for identity verification.
- Servo motor for gate/door simulation.
- LEDs for access indication.
- A central web dashboard was proposed to monitor and display RFID access events and patient data.

2. Hardware Integration

The hardware components were connected and tested in stages:

- The RC522 RFID Reader was connected to the ESP32 via SPI protocol.
- RFID cards and key fobs were assigned unique patient/staff IDs.
- The servo motor was connected to simulate gate opening when access was granted.
- LEDs were used to indicate access status (green for allowed, red for denied).
- A regulated power supply ensured stable operation of all components.
- PIR Sensor were connected for motion sensor.
- LCD were connected for displaying the process.

Each component was tested individually and then integrated into a single working prototype, mounted on a board for easy demonstration.

3. Embedded Programming

Using Arduino IDE, the ESP32 was programmed in Embedded C.

The logic included:

- Reading RFID UID from the reader.
- Verifying UID against predefined entries in memory.
- Controlling the servo motor to open the gate for authorized users.
- Triggering LEDs and serial print statements for feedback.
- Sending UID data via serial communication to the computer.

This code formed the heart of the embedded system controlling access operations in real-time.

4. System Testing and Validation

After development, the system underwent several testing phases:

- **Functional Testing:** Each RFID tag/card was tested to ensure proper UID reading and matching.
- **Access Testing:** Authorized tags opened the gate and displayed details; unauthorized tags were blocked.
- **Communication Testing:** Verified real-time serial data exchange between ESP32 and the dashboard.
- **Power Testing:** The system was evaluated under varied input voltages to test power regulation.

Each phase helped refine the logic and user interface to ensure accuracy, stability, and responsiveness.

5. Deployment Setup

The finalized setup included all components mounted neatly for demonstration. Wires were routed properly, and a clean interface was shown on the screen. Photographs of this implementation have been embedded in the previous section of this chapter for visual understanding.

ALGORITHM

Step 1: Start the system and initialize ESP32, RFID reader, and web interface.

Step 2: Scan the RFID card/tag when a patient, staff, or equipment is detected.

Step 3: Read the Unique RFID ID from the card/tag.

Step 4: Check the scanned ID in the hospital database.

Step 5: • If patient ID is found → Display patient details on the web and log the entry.

- If staff ID is found → Verify access permissions and control the entry-exit gate using a stepper motor.

- If equipment ID is found → Update the tracking status in the system.

- If ID not found → Deny access and trigger a buzzer alert.

Step 6: Send the scanned data to the web-based system for real-time monitoring.

Step 7: Update the database with the latest entry logs, staff movements, and equipment tracking.

Step 8: Repeat steps 2 to 7 for continuous monitoring.

Step 9: Stop the system when manually turned off.

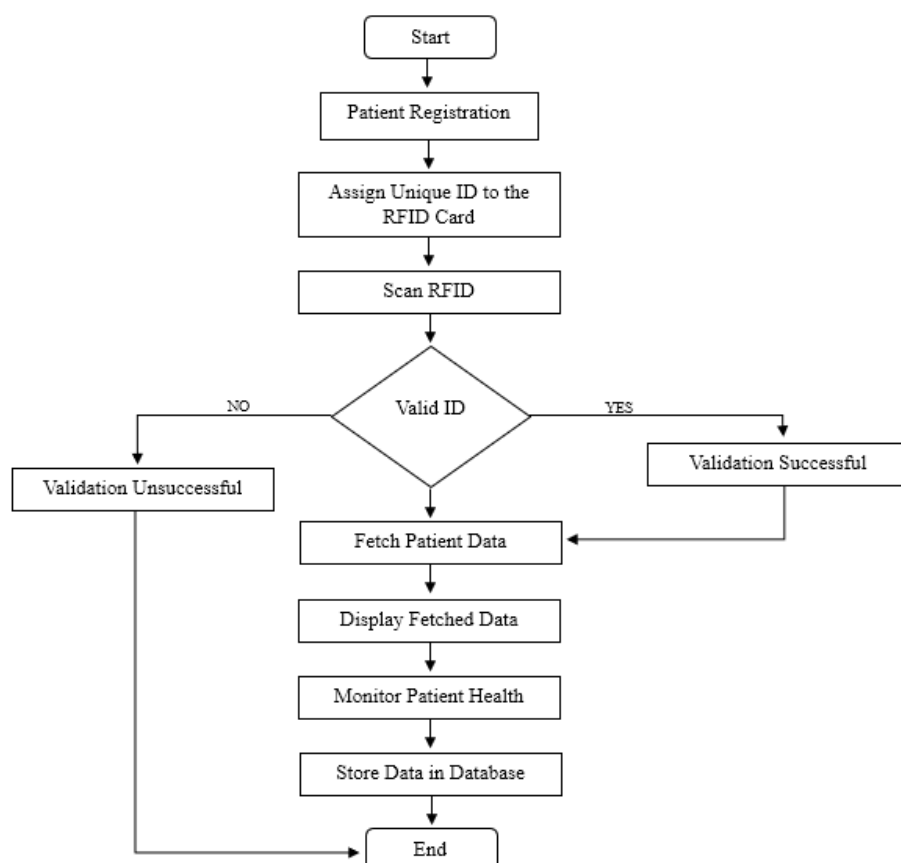


Figure: Flowchart

HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Description

1. ESP32



Figure: ESP32 Microcontroller

The ESP32 is a highly versatile and powerful 32-bit microcontroller developed by Espressif Systems. It's one of the most popular choices for developing IoT applications due to its built-in Wi-Fi and Bluetooth capabilities, which allow devices to communicate wirelessly. The ESP32 features a dual-core processor, which is capable of handling multiple tasks simultaneously, making it suitable for applications that require real-time processing and multitasking.

The microcontroller is also known for its flexibility, as it supports a wide range of communication protocols like I2C, SPI, UART, and PWM. These features allow it to interface with various peripherals and sensors seamlessly, making it a perfect choice for complex applications like smart systems and IoT devices. ESP32 modules come with various integrated peripherals, including digital-to-analog converters (DAC), analog-to-digital converters (ADC), and Pulse Width Modulation (PWM) capabilities, giving developers flexibility in interfacing with sensors, motors, and displays.

The ESP32 is well-suited for use in low-power IoT applications, thanks to its power management features, such as deep sleep mode (which consumes as low as 10 μA of current). This makes it ideal for battery-operated projects or systems where energy efficiency is a priority. With integrated hardware encryption and secure boot options, the ESP32 offers a high level of security, which is critical for IoT devices, especially in healthcare applications where privacy and data integrity are vital.

2. RFID Reader (RC522)



Figure: RFID Reader (RC522)

The RC522 is a compact, low-cost 13.56 MHz RFID module designed for reading RFID tags that adhere to the ISO/IEC 14443A standard. It is based on the MFRC522 chip, which supports a wide variety of RFID tags and smart cards. The module works by generating an electromagnetic field that powers the RFID tags and allows them to communicate with the reader. This contactless communication makes the RC522 module ideal for applications such as access control systems, inventory tracking, and asset management.

The RC522 module operates on the 13.56 MHz frequency, which is a global standard for RFID technology. This frequency is used for various applications such as personal identification, contactless payments, and ticketing systems. The RC522 can communicate with tags in close proximity (typically within 3-5 cm), which makes it useful in environments where security and quick access are required. The module is compact and easy to integrate with microcontrollers like the ESP32 via the SPI interface, which offers fast communication speeds for real-time operations. One of the key advantages of the RC522 is its low power consumption. It can be powered by a 3.3V or 5V power supply, and its low current draw makes it suitable for battery-operated systems. Additionally, the module is highly affordable, which makes it a popular choice for DIY projects and commercial applications. Its read-write capabilities allow it to interact with various types of RFID tags, from simple MIFARE cards to more complex RFID tags used in asset tracking or healthcare management applications.

3. RFID Passive Inlays

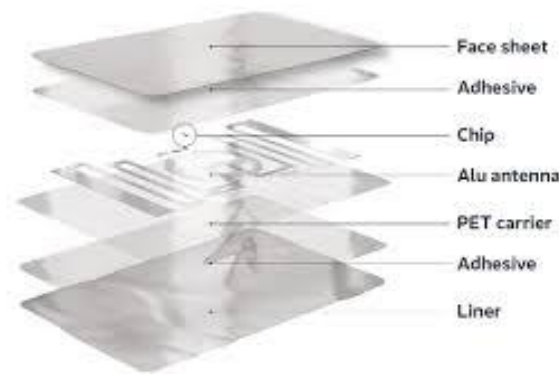


Figure: RFID Passive Inlay

RFID passive inlays are ultra-thin components made of an integrated circuit (IC) attached to a copper or aluminium antenna, all mounted on a flexible substrate (usually PET). These inlays are typically embedded within labels or inserted into plastic products to convert them into smart tags. Being "passive," they do not contain a battery but are powered by the electromagnetic field of the RFID reader. Passive inlays are the core of any RFID tag product they are highly customizable and used in applications ranging from supply chain management and retail to healthcare and pharmaceuticals.

4. RFID Card



Figure: RFID Card

RFID cards are smart cards with an embedded passive RFID inlay inside a PVC shell, commonly used for access control, identity verification, and transaction processing. They are the size of a credit card and can be easily carried in a wallet. These cards operate using radio waves and can be scanned quickly, even without direct line-of-sight, making them efficient and secure in high-traffic environments like hospitals, offices, and campuses. Most RFID cards are rewritable and can store user-specific data.

5. RFID Key Fob



Figure: RFID Key fob

RFID key fobs are small, durable tokens that contain the same electronics as RFID cards but in a rugged plastic case designed to hang on keyrings. These are particularly useful in applications requiring frequent handling, portability, and durability. Their ergonomic design and sturdy build make them ideal for long-term use by patients and visitors. The fob contains an embedded coil and chip, just like cards, but often in a rounded or teardrop shape.

6. Servo Motor (SG90)

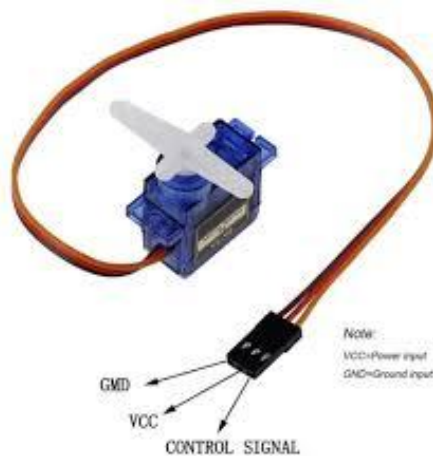


Figure: Servo Motor (SG90)

The SG90 is a small and lightweight servo motor designed for precise control over angular movement. It has a range of motion between 0° and 180° and can be controlled via a PWM signal, allowing it to rotate to specific angles. Servo motors are commonly used in applications requiring high accuracy and quick response, such as robotics, automated systems, and access control systems.

The SG90 is a high-precision servo motor that is often used in small-scale projects and IoT devices due to its compact size and lightweight design. It is powered by a 5V supply and can be controlled using PWM signals from microcontrollers like the ESP32.

7. Power Supply

Power supply is necessary for providing stable voltage and current to the system components. In embedded systems like the Hospital Management System, a 5V DC power supply is used to power components like the ESP32, RC522 RFID reader, servo motor, and LEDs. The power supply ensures that each component operates correctly and efficiently, preventing voltage fluctuations that could lead to instability or failure.

SOFTWARE DESCRIPTION

1. Arduino IDE

The Arduino IDE (Integrated Development Environment) is an open-source application designed for writing, compiling, and uploading embedded C/C++ code to Arduino and compatible microcontrollers. It simplifies hardware-software interaction and is widely used in academic, hobbyist, and prototyping environments. Arduino IDE supports embedded C/C++ programming and includes a simple code editor with features such as syntax highlighting, automatic indentation, and error notifications. It allows users to manage libraries, interface with hardware through built-in functions, and monitor serial communication in real-time using the Serial Monitor. The IDE provides support for third-party board definitions, enabling it to work with a range of microcontrollers including the ESP32. The integrated bootloader simplifies firmware uploading over USB, while the vast online community and library support allow for quick integration of modules like RFID, Servo, and SPI.

In our project, the Arduino IDE is used to develop and upload the control program to the ESP32 board. The embedded code handles RFID tag reading using the MFRC522 module, servo motor control for gate access, LED status indication, and serial communication with the PC. The Serial Monitor is utilized during development for real-time debugging of tag reads and logic flow.

2. Visual Studio Code

Visual Studio Code (VS Code) is a lightweight, yet powerful source code editor developed by Microsoft. It supports a wide array of programming languages and frameworks and is highly extensible through its marketplace plugins. It is especially popular among developers for Python, web development, and embedded systems integration. VS Code offers a highly customizable development environment with

features such as IntelliSense (smart code completion), syntax highlighting, Git integration, terminal access, and debugging tools. It supports extensions for Python, HTML, CSS, and serial port monitoring, making it ideal for both backend and frontend development. Real-time collaboration is also possible via Live Share. With Python extensions, VS Code enables execution of scripts for serial communication, data logging, and GUI updates. The editor's file management and workspace feature also make it convenient for handling multi-file projects. In our hospital management system, VS Code is used as the central development platform for both backend Python code and the frontend HTML/CSS interface. Python scripts written here read serial data from the ESP32, identify RFID tags, and update hospital access logs. The HTML and CSS code, also written in VS Code, forms the user dashboard interface displaying login information and patient data dynamically based on tag reads.

3. HTML (HyperText Markup Language) and FLASK

HTML (HyperText Markup Language) is the standard language used to design and structure web pages. It is used to create the layout of a website by defining elements such as headings, paragraphs, tables, forms, and buttons. In this project, HTML is used to develop the user interface of the hospital management system, where patient details, appointment information, doctor data, and other records are displayed in a clear and organized manner. It provides a simple and user-friendly way for users to interact with the system.

Flask is a lightweight web framework in Python that is used to build web applications and handle the backend operations. It acts as a bridge between the frontend (HTML) and the database. In this project, Flask is used to process the data received from the system, retrieve information from the database, and send it to the HTML pages for display. It also handles routing, user requests, and data processing. By combining HTML and Flask, the system is able to provide a dynamic and interactive hospital management platform where data can be accessed, updated, and managed efficiently.

SYSTEM DESIGN

Block Diagram

BLOCK DIAGRAM

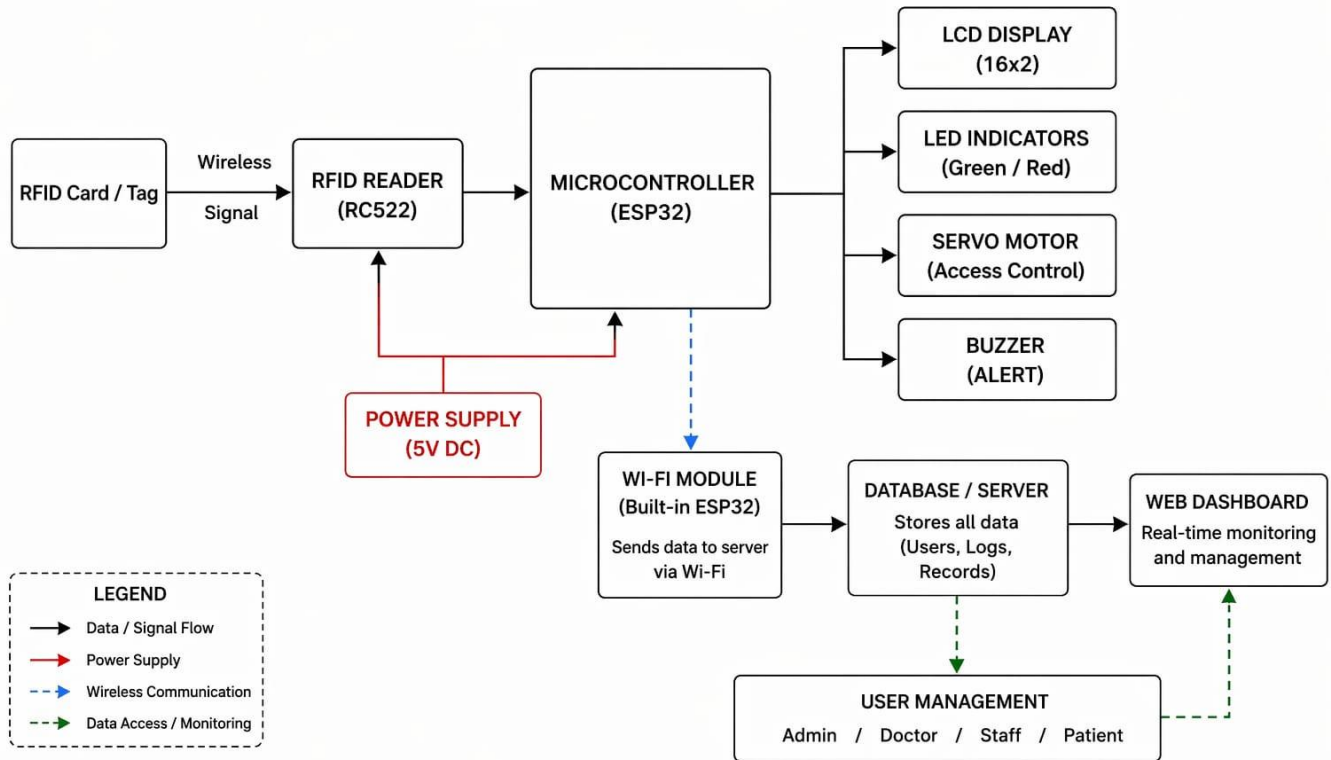


Figure: Block Diagram

Proposed Solutions

The Hospital Management System using RFID is designed to overcome inefficiencies in traditional hospital operations by integrating RFID technology, an ESP32 microcontroller, and a web-based monitoring platform. Each patient is assigned an RFID card, enabling automated identification and instant access to medical records, reducing manual paperwork and errors. Similarly, hospital staff will use RFID-enabled ID cards for secure authentication and controlled access to restricted areas, managed through a stepper motor-controlled entry-exit system. To prevent misplacement and optimize resource utilization, RFID tags will be attached to medical equipment, allowing for real-time tracking and theft prevention. Additionally, the system will automate medicine and inventory management, ensure timely restocking and minimize supply shortages. The integration of ESP32 facilitates wireless communication with a web-based hospital management system, providing real-time data on patient records, staff movements, and medical equipment tracking.

By implementing this system, hospitals can enhance security, improve operational efficiency, reduce human errors, and ensure smooth workflow management. The solution offers a cost-effective and scalable approach to modernizing hospital operations through RFID-based automation and digital monitoring.

Implementation

A prototype model was developed to validate the Hospital Management System using RFID technology. The setup focuses on patient data management, staff access control, secure pharmacist access to prescriptions, and equipment data management. The central component of the system is the ESP32 microcontroller, which interfaces with three RC522 RFID readers via the SPI communication protocol. These readers are responsible for scanning RFID tags on patients, staff, and equipment. The servo motor is used for staff access control, unlocking doors when authorized RFID tags are detected. The RFID system also handles patient data management by tracking patient identification and medical history stored on RFID tags. For secure pharmacist access to prescriptions, only authorized RFID tags allow access to prescription data, reducing the risk of medication errors. Equipment data management is achieved by tagging hospital assets with RFID, enabling real-time tracking of equipment location and usage. The system is connected to a laptop hosting the web application interface, which allows real-time monitoring of patient data, staff access, prescription logs, and equipment status. The application provides an interactive platform for managing hospital operations efficiently and securely.

RESULTS

The system was rigorously tested under various scenarios to evaluate its performance. The RFID readers successfully identified authorized tags, leading to servo motor actuation and granting access. Unauthorized tags were denied access, ensuring system reliability and security.

The developed web dashboard efficiently captured and displayed real-time data including staff login credentials, patient information, and entry-exit logs.

Key observations include:

- Successful RFID authentication within 1–2 seconds of scanning.
- Real-time updating of staff attendance and patient data.
- Reliable servo motor actuation for entry control without noticeable delay.
- Pharmacists were provided with RFID-based access to patient prescriptions through the web dashboard.
- Equipment data was managed through RFID tags, indicating whether a device was operational or under maintenance.

FUTURE SCOPE

The proposed Hospital Management System using RFID Technology offers several opportunities for future enhancement and scalability:

- **Cloud Integration:** Future versions can integrate cloud-based storage to allow secure, centralized access to patient records and logs across multiple hospital branches.
- **Mobile Application Support:** A dedicated mobile app can be developed for staff and administrators to view real-time data, receive alerts, and manage hospital operations remotely.
- **Biometric Authentication:** Combining RFID with biometric verification (e.g., fingerprint or facial recognition) can further strengthen access control and identity validation.
- **AI-Based Predictive Maintenance:** Using AI algorithms, equipment data can be analyzed to predict maintenance needs before failures occur, improving hospital efficiency and safety.
- **Integration with National Health Portals:** Linking with government or insurance health databases could streamline patient record management and facilitate automated billing.

CONCLUSION

The RFID-based Hospital Management System developed in this project provides an effective solution to improve patient identification, access control, and data management in hospitals. By assigning unique RFID IDs to patients, doctors, staff, and administrators, the system ensures accurate and fast identification, reducing the chances of human errors associated with manual processes.

The integration of a centralized database and user-friendly dashboard enables quick retrieval and monitoring of important information such as patient details, appointments, and doctor data. This improves workflow efficiency and helps healthcare professionals make faster and better decisions.

The system also enhances security by allowing only authorized users to access specific data, ensuring proper data handling and privacy. Although the project is implemented at a prototype level, it clearly demonstrates the potential of RFID technology in modernizing hospital operations.

Overall, the project successfully shows how the use of RFID technology can create a more efficient, reliable, and user-friendly hospital management system, ultimately contributing to improved healthcare services and patient satisfaction.